

ORYAHOVO, Bulgaria

WMO: 155140

Lat: 43.683N

Lon: 23.967E

Elev: 31

StdP: 100.95

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-8.2	-12.3	1.3	-10.1	-9.8	1.6	-7.4	19.7	3.0	16.3	3.0	2.3	90	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	33.7	22.7	32.0	22.3	30.6	21.7	24.1	31.2	23.1	30.1	22.3	29.1	2.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.5	28.4	20.7	15.5	27.2	19.9	14.6	26.3	72.5	31.2	68.6	30.2	65.4	29.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	15.7	11.7	-12.9	37.1	3.3	2.5	-15.3	38.9	-17.3	40.4	-19.1	41.8	-21.5	43.6		
(5)			-13.1	27.4	3.0	1.8	-15.2	28.7	-16.9	29.8	-18.6	30.8	-20.8	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-0.2	2.4	7.8	13.3	19.0	22.3	25.0	24.5	19.8	13.1	7.2	1.2		
	DBStd	9.63	4.99	5.39	4.62	3.26	3.28	3.18	2.74	2.76	3.35	3.97	3.85	4.26		
	HDD10.0	1031	318	218	97	9	0	0	0	0	0	16	101	273		
	HDD18.3	2601	576	447	327	155	33	6	1	1	24	167	334	531		
	CDD10.0	2123	1	4	28	107	278	368	464	450	293	113	17	1		
	CDD18.3	651	0	0	0	3	53	123	206	193	67	6	0	0		
	CDH23.3	5254	0	0	1	19	314	956	1874	1643	417	28	2	0		
	CDH26.7	1867	0	0	0	2	55	299	761	637	109	3	0	0		
(14) Wind	WSAvg	2.9	2.9	3.4	3.7	2.9	2.8	3.0	2.9	2.6	2.8	2.5	2.7	2.8		
(15) Precipitation	PrecAvg	562	38	32	41	48	66	60	56	39	52	51	36	43		
	PrecMax	934	89	70	93	121	114	122	123	196	151	108	98	105		
	PrecMin	328	7	2	9	2	15	12	11	6	4	5	0	2		
	PrecStd	137	22	19	24	25	31	27	36	39	38	33	24	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	17.1	21.7	25.8	30.3	34.2	36.9	35.9	32.0	26.3	20.4	14.9		
		MCWB	9.1	10.8	12.5	16.6	20.1	22.7	23.0	22.6	21.1	19.0	14.2	10.1		
	2%	DB	11.4	14.8	19.1	22.4	28.0	31.5	34.1	33.3	29.6	23.3	17.0	10.7		
		MCWB	7.6	9.3	11.3	14.9	19.2	22.2	22.9	22.6	19.9	17.1	12.8	7.9		
	5%	DB	9.0	12.3	16.9	20.5	26.5	29.9	32.3	31.7	27.4	21.1	14.7	7.9		
		MCWB	6.0	7.9	10.2	13.8	18.5	21.5	22.5	22.1	19.2	16.1	11.6	6.2		
	10%	DB	6.5	9.9	14.9	18.9	24.9	28.3	30.7	30.2	25.6	19.1	12.8	6.5		
		MCWB	4.5	6.5	9.3	12.9	17.8	20.9	22.2	21.3	18.3	14.9	10.5	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	11.2	13.5	17.5	21.3	24.5	25.7	24.9	21.9	19.9	15.1	10.4		
		MCDB	13.7	16.5	19.7	24.1	28.2	30.6	32.0	32.1	30.3	25.1	19.2	14.3		
	2%	WB	7.7	9.6	12.0	15.7	20.2	23.2	24.4	23.6	20.7	17.9	13.3	8.2		
		MCDB	11.0	14.0	17.7	21.2	26.4	30.3	31.3	31.1	27.9	22.4	15.8	10.3		
	5%	WB	6.3	8.1	10.7	14.5	19.2	22.2	23.4	22.7	19.8	16.5	12.2	6.6		
		MCDB	8.8	11.6	16.3	19.3	25.3	28.7	30.4	29.7	26.3	20.3	14.4	7.7		
	10%	WB	4.6	6.9	9.6	13.5	18.4	21.4	22.4	21.9	18.8	15.2	10.9	5.2		
		MCDB	6.4	9.7	14.4	17.9	23.8	27.3	29.3	28.9	24.6	18.5	12.6	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.9	7.9	8.6	9.1	9.0	9.7	9.8	9.3	7.5	5.3	4.2		
		MCDBR	8.2	10.2	12.0	11.4	11.5	11.5	12.2	12.1	12.4	10.5	9.0	6.6		
	5% WB	MCWBR	5.8	6.5	6.4	5.7	5.2	4.9	5.4	4.7	5.6	5.8	5.5	4.8		
		MCWBR	7.6	9.6	11.4	10.3	10.4	10.5	10.8	10.9	11.5	9.8	7.3	6.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.375	0.415	0.474	0.453	0.453	0.464	0.479	0.446	0.404	0.371	0.348		
		taud	2.294	2.216	2.161	2.013	2.146	2.186	2.159	2.114	2.172	2.278	2.325	2.318		
	Ebn at Noon	Ebn at Noon	759	798	813	792	824	824	809	780	773	758	724	717		
		Edh at Noon	86	110	132	166	149	144	146	148	130	103	84	77		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.45	2.29	3.43	4.69	5.75	6.39	6.48	5.77	4.18	2.71	1.55	1.20		
	RadStd	RadStd	0.22	0.31	0.37	0.50	0.48	0.44	0.40	0.42	0.33	0.35	0.22	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.49	N/A	N/A	N/A	+0.40	N/A	N/A	-141	+70	N/A

Nomenclature: See separate page